Personal Learning and Work Log

Name: Marvin Oliver Tan (陈威宏)

Student ID: 1820262139

Team Name: PentaByte

Project: Algorithms, Data Structures and Connectionist AI

Course: Foundations of Computer Science & AI

Stage: 2

Reporting Period: From the beginning of course learning and project work to October 24, 2026

October 29, 2026 — Create Live Status Report PPT

Activity and Purpose

Create a PPT presentation slide for the live status report.

My Contribution

I'm responsible for filling in the sections of team introduction, project direction, and the representative example as evidence for the project progress. I'm also tasked to present the respective parts during the live status report, hence I practiced explaining the slides.

Outcome

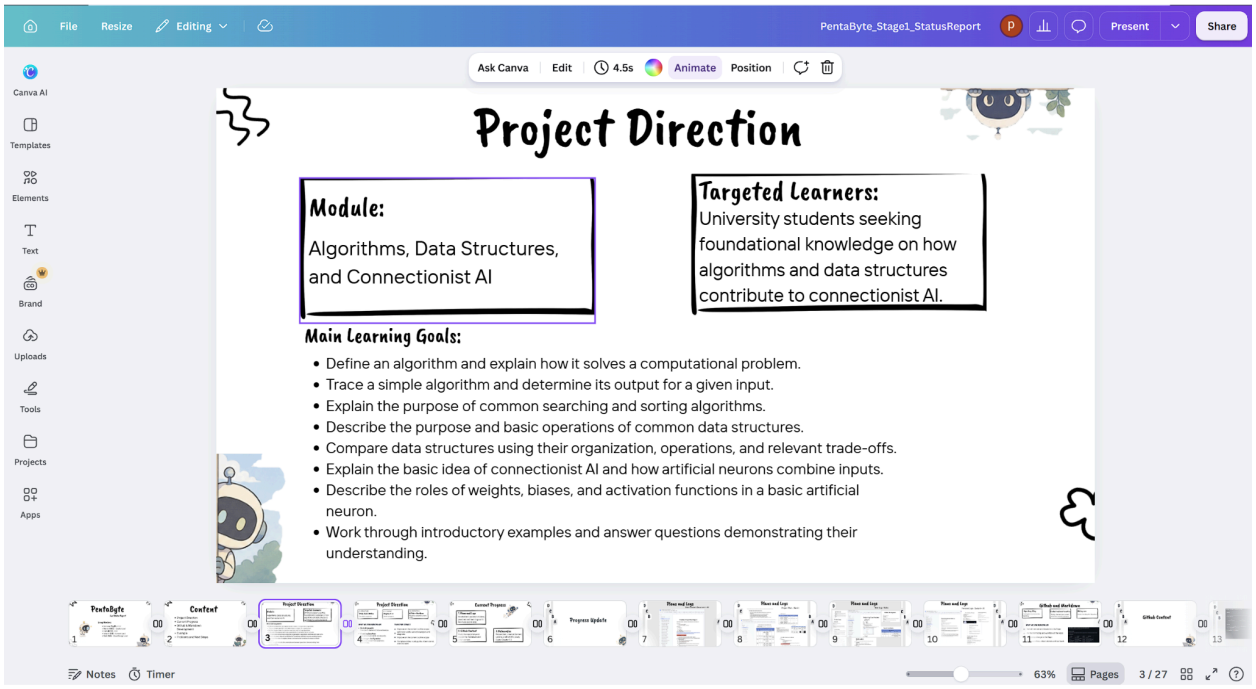
I created the slides for project progress and the representative example and ensured the presentation slide is ready for the live status report. I also practiced and understood what I needed to present during the live report.

Learning and Reflection

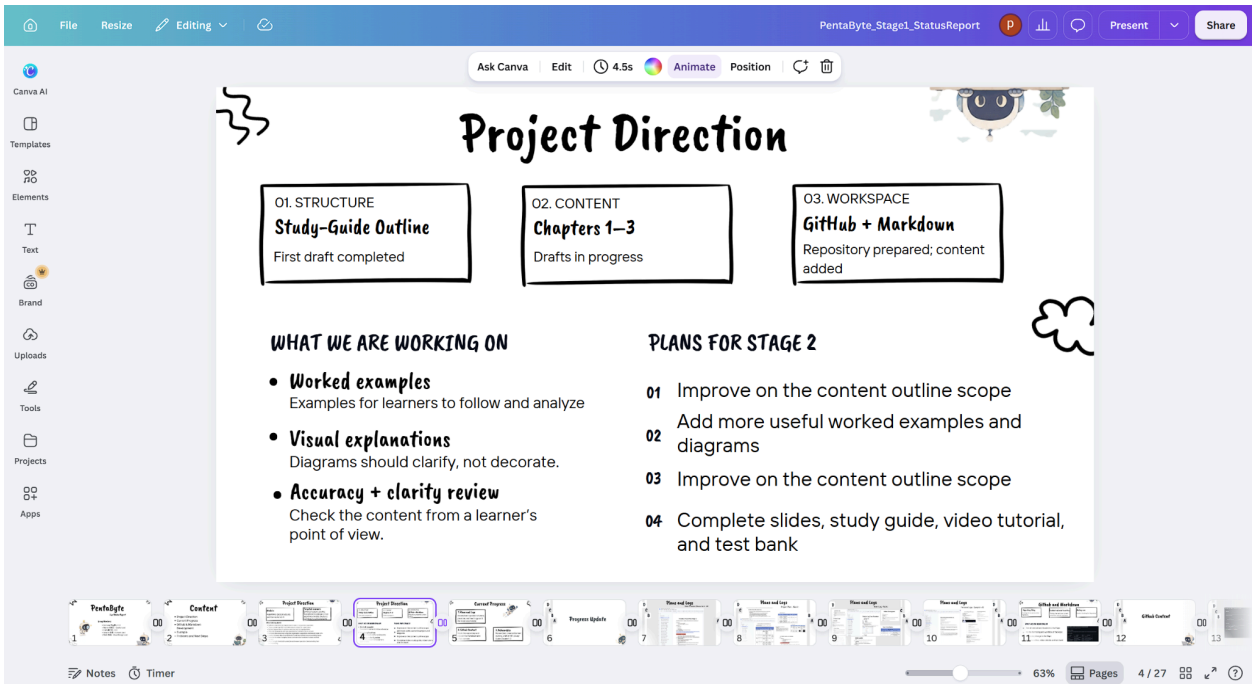
Throughout the process of creating the presentation slides, I was unsure of how I would approach presenting our status report in front of class, since I wasn't aware of the expectations placed on us. However, after revisiting the assignment instruction page, I noticed several details I missed

that I used to complete the slides. This made me realize that I should pay deep attention to assignment details to not miss anything important that might deduct my scores.

Proof



Screenshot of project direction part (1)



Screenshot of project direction part (2)



Creating the PPT with Nicholas (吴宜展）

AI Use

None

Next Steps

Present the PPT during class

October 30, 2026 — Present Live Status Report + Planning

Activity and Purpose

Prepare and present the live status report of our team.

My Contribution

I was tasked to present the slides that I prepared (project direction, team introduction, and representative example). Beforehand, I practiced presenting the slides on my own to build confidence and familiarity with the status report.

Outcome

The status report went quite well in my opinion. Not only were we able to present under the 4 minute restriction, we also managed to present our ideas in a coherent manner. The audience seemed more engaged than I was expecting thanks to the hook made by Nicholas (吳宜展)

Learning and Reflection

After the presentation, I realized that I needed to improve more on my communication skills, since I did stutter quite a bit despite rehearsing a lot beforehand. I also noticed that the project style is more catered towards an entertaining style rather than a formal and overly academic tone, which sets the plan for our next step which is to create the deliverables.

Proof

[Class live status report schedule as arranged by the teacher]

AI Use

None

Next Steps

Hold a meeting to brief about the next steps for stage 2

October 31, 2026 — Meeting + Next Steps Planning

Activity and Purpose

Discuss and plan for stage 2.

My Contribution

Recap the study guide assignment, start the study guide template,

Outcome

Planned the next course of action and work plan for this week. Workload is separated into the 5 units, with each member covering one unit.

Learning and Reflection

I learned what to do for each deliverable and recapped the overall workflow of the project.

Proof



Group Picture during Meeting



Elevator Picture

AI Use

None

Next Steps

Continue deliverable production.

November 03, 2026 — Understand Unit 5 + Testing ChatGPTplus

Activity and Purpose

Understand the content outline for unit 5 and start structuring a study guide format.

My Contribution

I started to create a study guide cover page and table of content design. While doing so, I also focused on understanding the content of unit 5

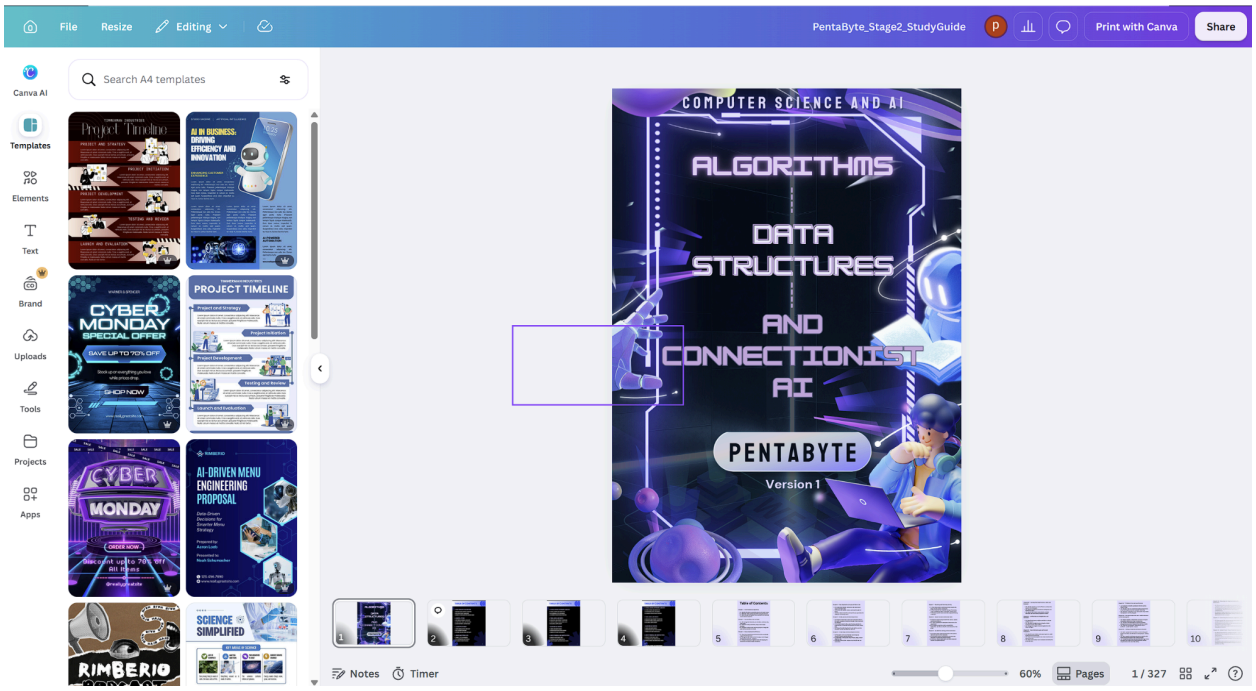
Outcome

Created a cover page as shown in the “proof” section and understood a bit of the content in unit 5.

Learning and Reflection

While some of the team’s members gathered together, I worked on my own task of trying to construct the cover page design for the study guide. While I did succeed making the cover page, in the time we had, I wasn’t able to complete the table on content. This led me to realizing that if I made the study guide manually, I won’t be able to complete the task in time. This led me to AI use, where Nicholas (吳宜展) learned the scope of ChatGPT’s capabilities. From this I wonder how I can use AI to help make the study guide format and content. For now, I will consider this opportunity to learn how to use AI effectively to reduce workload that would normally take a long time to complete.

Proof



Planning the design of the study guide (cover page)



Picture of some members of the team working together

AI Use

None

Next Steps

Find out how to use AI to format a study guide as an example to follow.

November 05, 2026 — Finalizing Content Unit 3

Activity and Purpose

Learn how to use AI to create a study guide format and generate prompts for it to generate the study guide format.

My Contribution

During the afternoon, I tried several prompts for ChatGPT to produce a study guide format that I can follow. However, the output wasn't satisfactory, as it generated a format that didn't quite meet my expectations. By the evening, the team got together for a meeting. I was responsible for continuing to generate format ideas for the study guide. Despite this, I keep getting outputs which didn't fit the improvement standards that I asked for. Hence, I turned to Nicholas (吳宜展), who studied how to use AI and how to prompt engineer, for advice on how to prompt well so the AI gives me a good result.

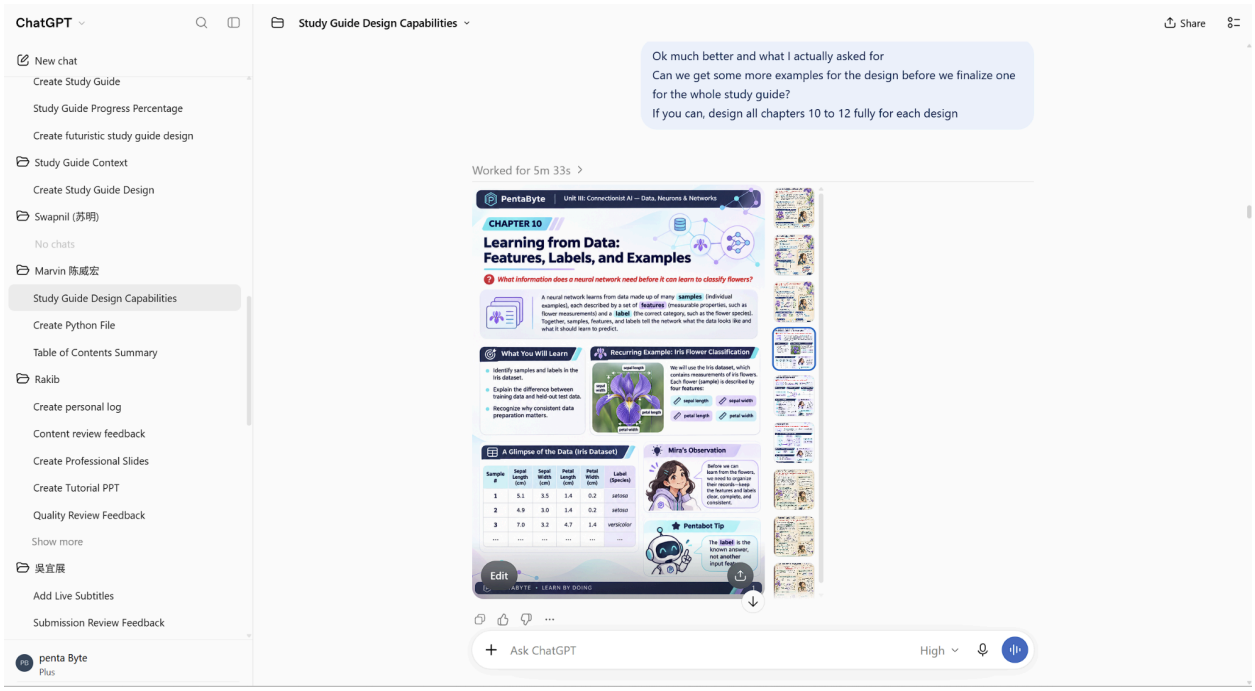
Outcome

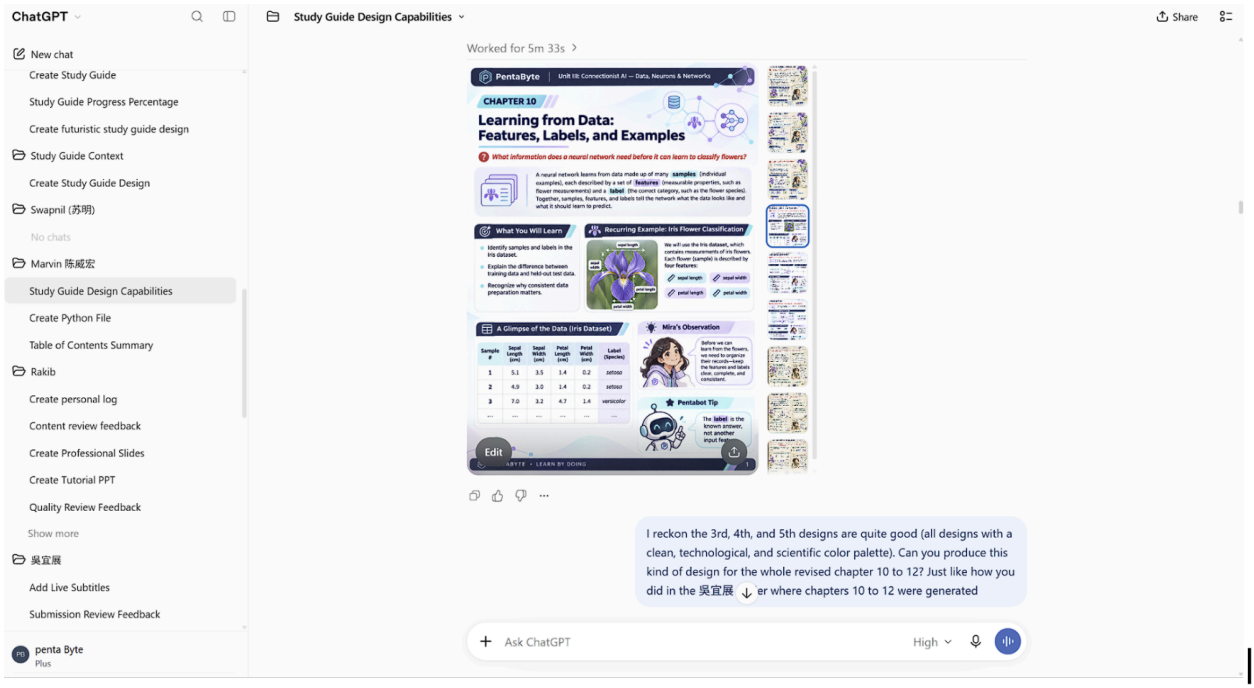
Found out about prompt engineering and got a baseline on how to create a study guide format using AI.

Learning and Reflection

By the end of the day, I was disappointed due to the AI not responding anymore. The apparent problem was a connection issue, however I couldn't locate where the problem came from. Even then, what AI did generate is either a plain design for a complete study guide or an image for only one page, neither of which is what I want. Despite this, I did go through a lot of trial and error to refine my prompting skills and gained new information from the team trying to learn how to prompt and ask AI exactly what to do.

Proof





Unsatisfactory ChatGPT outputs


XIAOMI 11T | BIT 2026
05/11/2026 23:55

Getting help on prompt engineering/how to use ChatGPT effectively



Held a meeting and discussed project updates.

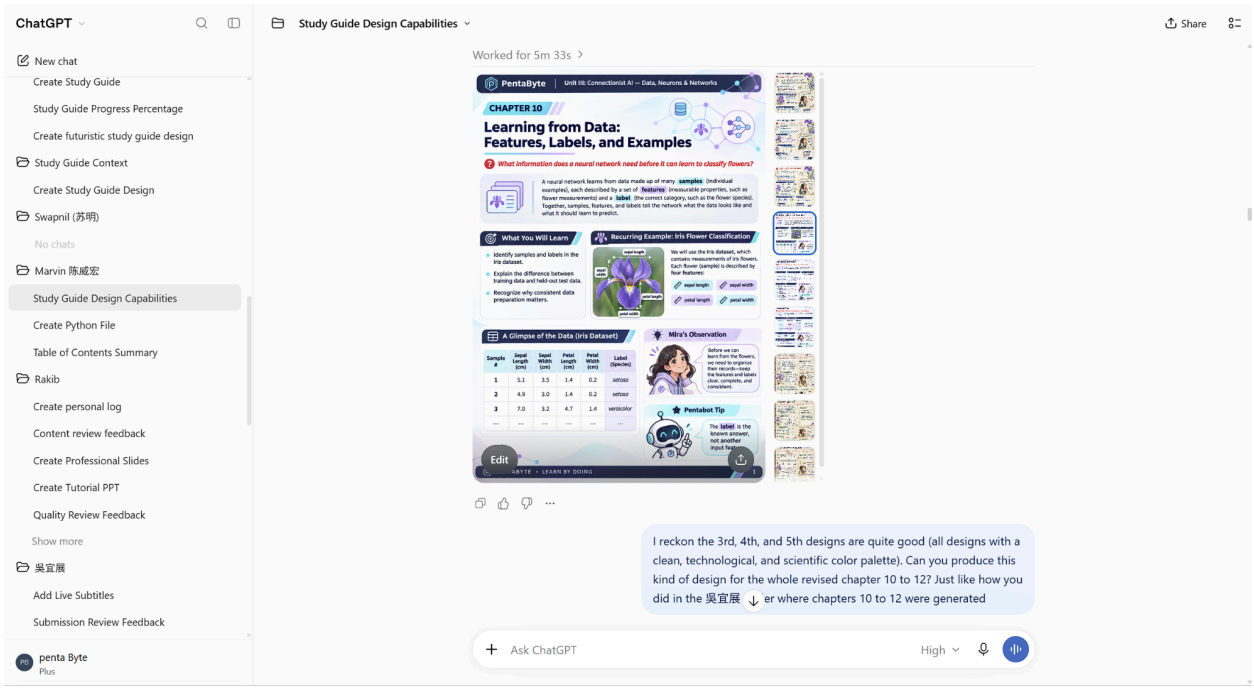
AI Use

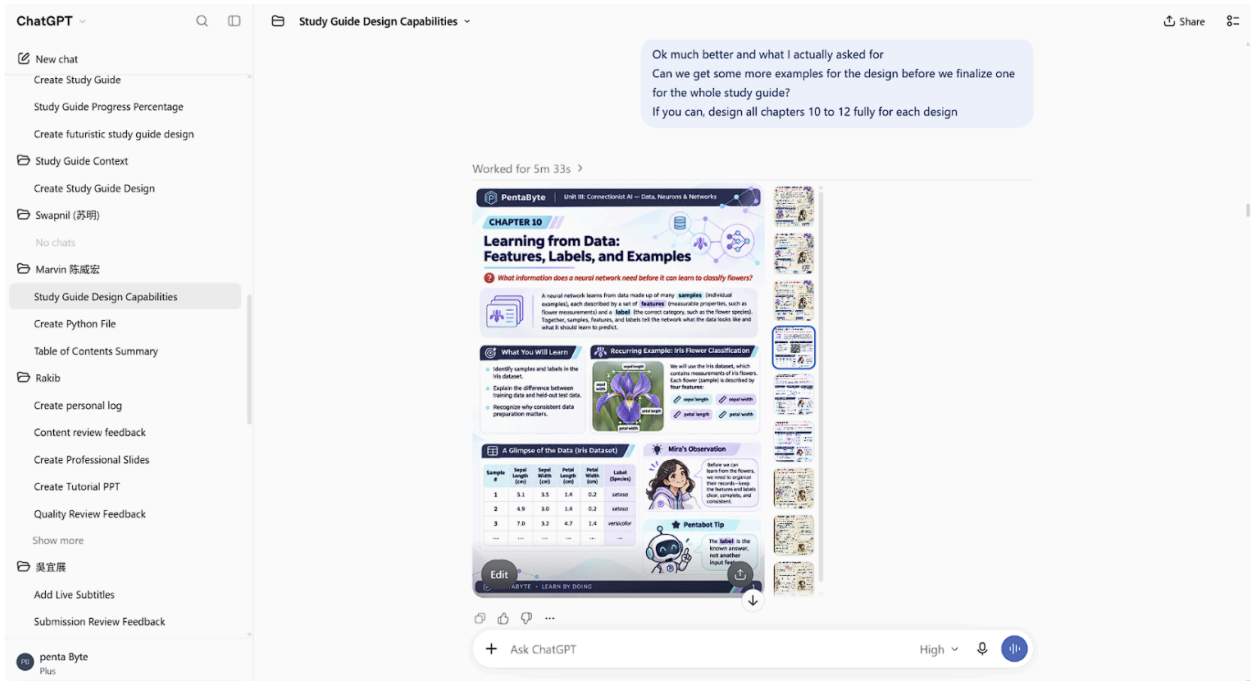
Tool: ChatGPT.

Purpose: Create the study guide format.

Verification: Check the study guide requirements.

Prompt with screenshot:





Next Steps

Continue making the study guide format using the help of AI.

November 06, 2026 — Update Personal Log, Finalize Study Guide Format, Unit 5 PPT.

Activity and Purpose

Start to finish the deliverables.

My Contribution

I did a lot of things on this day, such as updating my personal log to the latest activity, prompting ChatGPT to create a finalized study guide, and completing the unit 5 presentation slides.

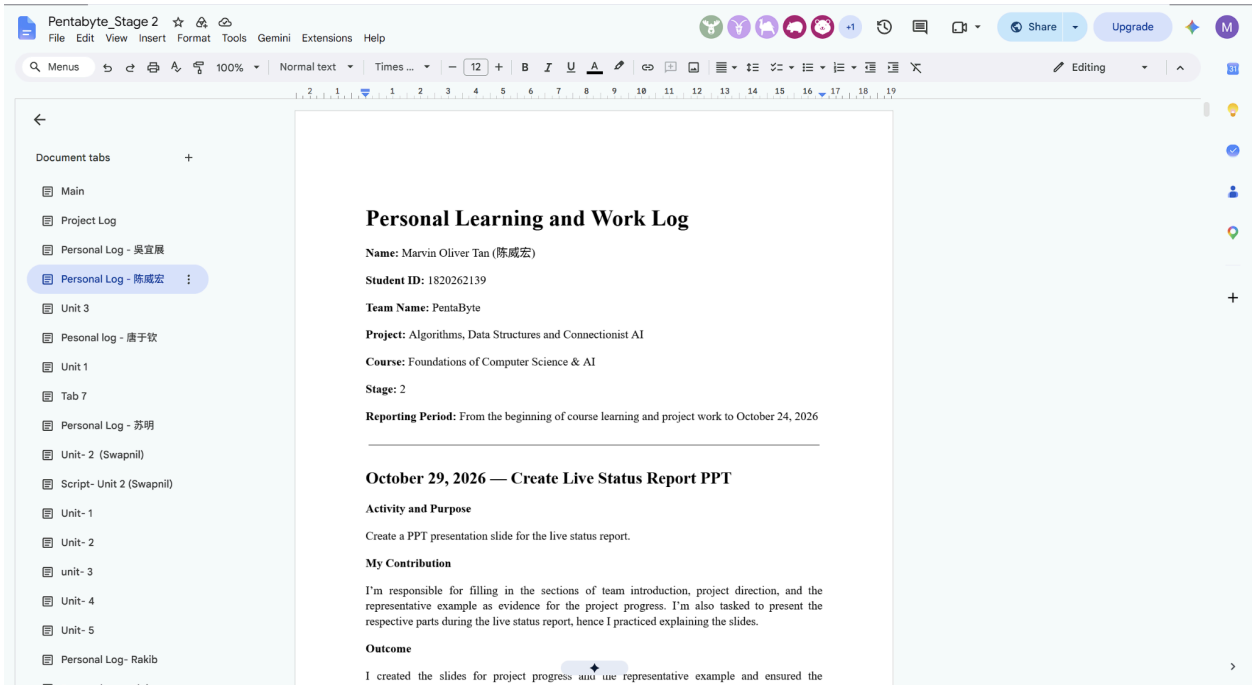
Outcome

A completed study guide format, updated personal log, and a PPT draft for the recording the next day.

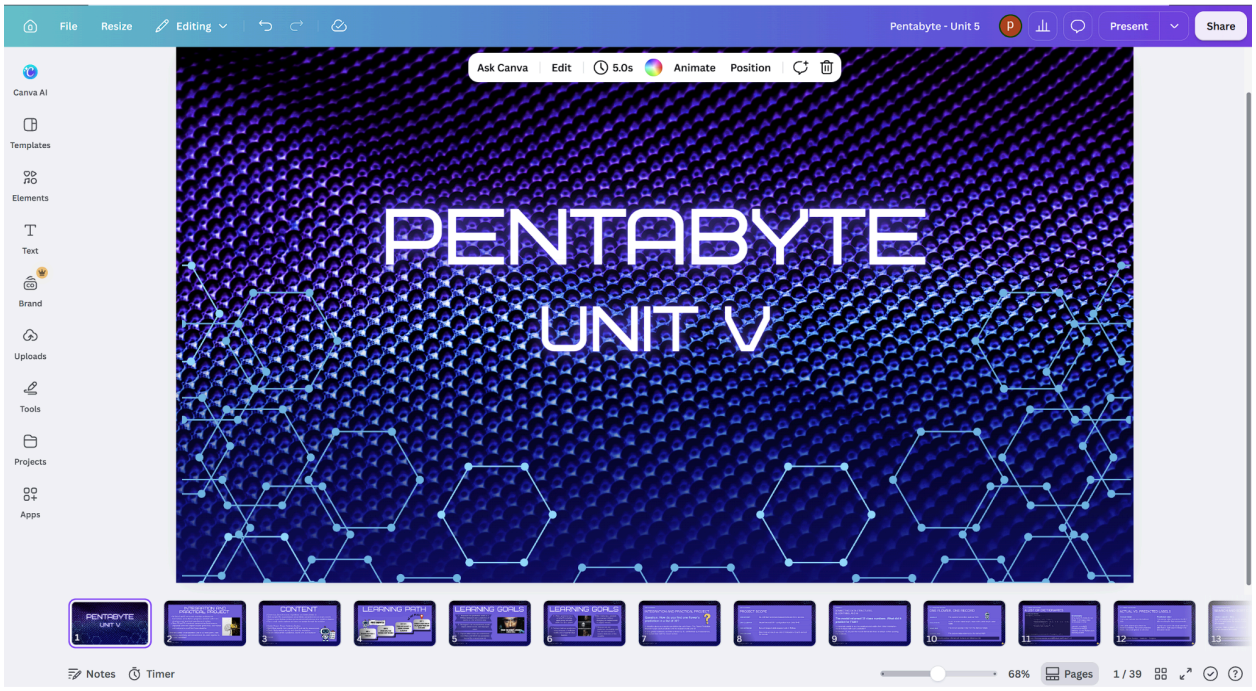
Learning and Reflection

I finally learned how to properly prompt engineer to get a decent study guide format that I can use and edit, while also completing other important parts of the assignment. This process has taught me that AI use is also a skill that is prominent in today's world. This is why I have to practice using AI for the foreseeable future to be able to reduce workload effectively.

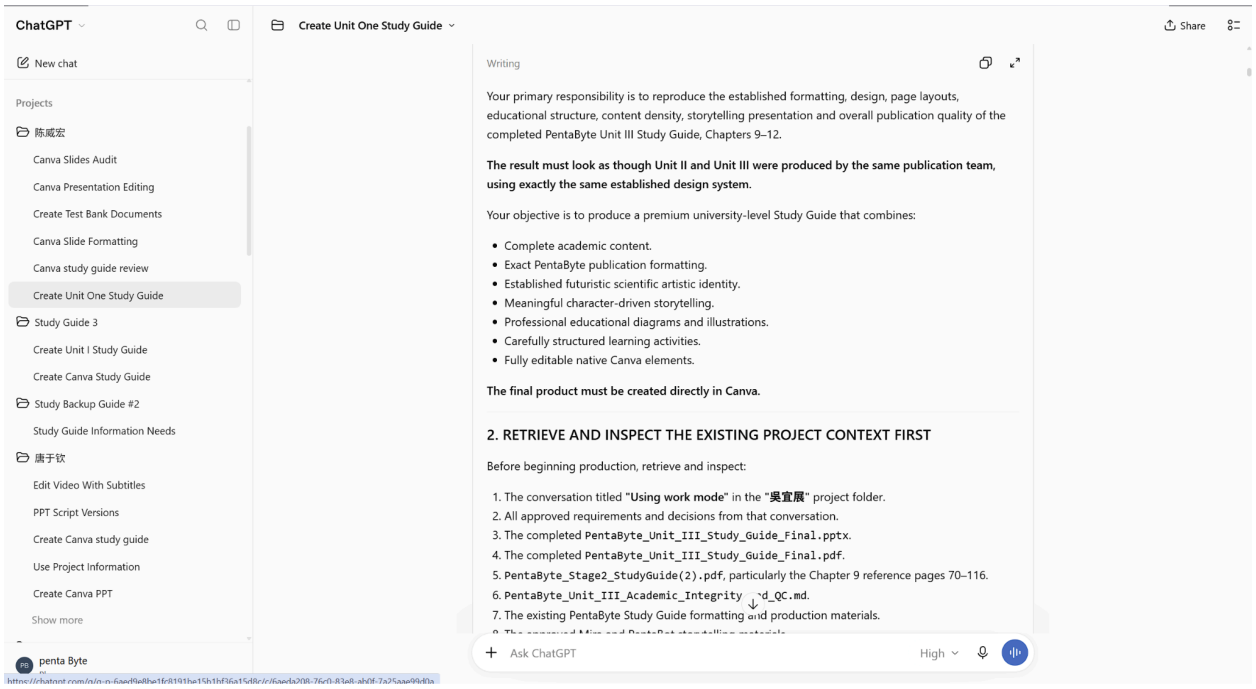
Proof



Updated Personal Log



Finalizing unit 5 presentation slide.



Prompt engineering with AI

AI Use

Tool: ChatGPT.

Purpose: Create a study guide format and PPT presentation slide

Verification: Check the produced product

Prompt and screenshot:

New chat

Projects
陈威宏
Canva Slides Audit
Canva Presentation Editing
Create Test Bank Documents
Canva Slide Formatting
Canva study guide review
Create Unit One Study Guide
Study Guide 3
Create Unit I Study Guide
Create Canva Study Guide
Study Backup Guide #2
Study Guide Information Needs
唐于钦
Edit Video With Subtitles
PPT Script Versions
Create Canva study guide
Use Project Information
Create Canva PPT
Show more

penta Byte
Plus

Yesterday 5:46 PM

Refer to the "Using work mode" chat in the "吳宜展" folder to extract both context for this chat and the standards that I want you to achieve in this chat. This is for making the study guide for unit 1 (editable and formatted properly based on the instructions given in the "Using work mode" chat)

Worked for 2m 55s

I've retrieved the relevant context from the **"Using work mode"** discussion in the **吳宜展** project. I'll use it as the production brief for this conversation.

The objective is to create:

PentaByte — Unit I Study Guide
Unit I: Computational Thinking and Python Foundations
Chapters 1–3.

This is not supposed to be a basic summary, a conventional textbook, or a collection of decorative slides. It must be a professionally designed, academically complete, beginner-friendly study guide with native editable elements and a consistent educational storyline.

The most important principle is:

Preserve the academic depth of the approved content while achieving the visual quality, layout consistency, and teaching effectiveness of the revised Unit III reference.

1. Context and standards carried forward

Area	Required standard

Ask ChatGPT High

ChatGPT
Create Unit One Study Guide
Share

New chat

Projects
陈威宏
Canva Slides Audit
Canva Presentation Editing
Create Test Bank Documents
Canva Slide Formatting
Canva study guide review
Create Unit One Study Guide
Study Guide 3
Create Unit I Study Guide
Create Canva Study Guide
Study Backup Guide #2
Study Guide Information Needs
唐于钦
Edit Video With Subtitles
PPT Script Versions
Create Canva study guide
Use Project Information
Create Canva PPT
Show more

penta Byte
Plus

Ok, before we proceed to the next step, any other details I should add for you to know EXACTLY what to do? (Remember use chapters 9 to 12 to generate unit 1 EXACTLY as the referenced chapters. I don't want any random newly generated design, format, content, and structure. I need EXACTLY what you did for chapters 9 to 12 for unit 1 for now)

Worked for 1m 19s

Honestly, **you don't need to add many more instructions.** I now understand the level of consistency you're expecting. However, I want to lock down a few important details before production so we don't repeat the problems from previous drafts.

I've also located the actual Unit III reference files in your project materials.

1. The reference materials I will treat as authoritative

I will use three levels of reference, not just Chapter 9.

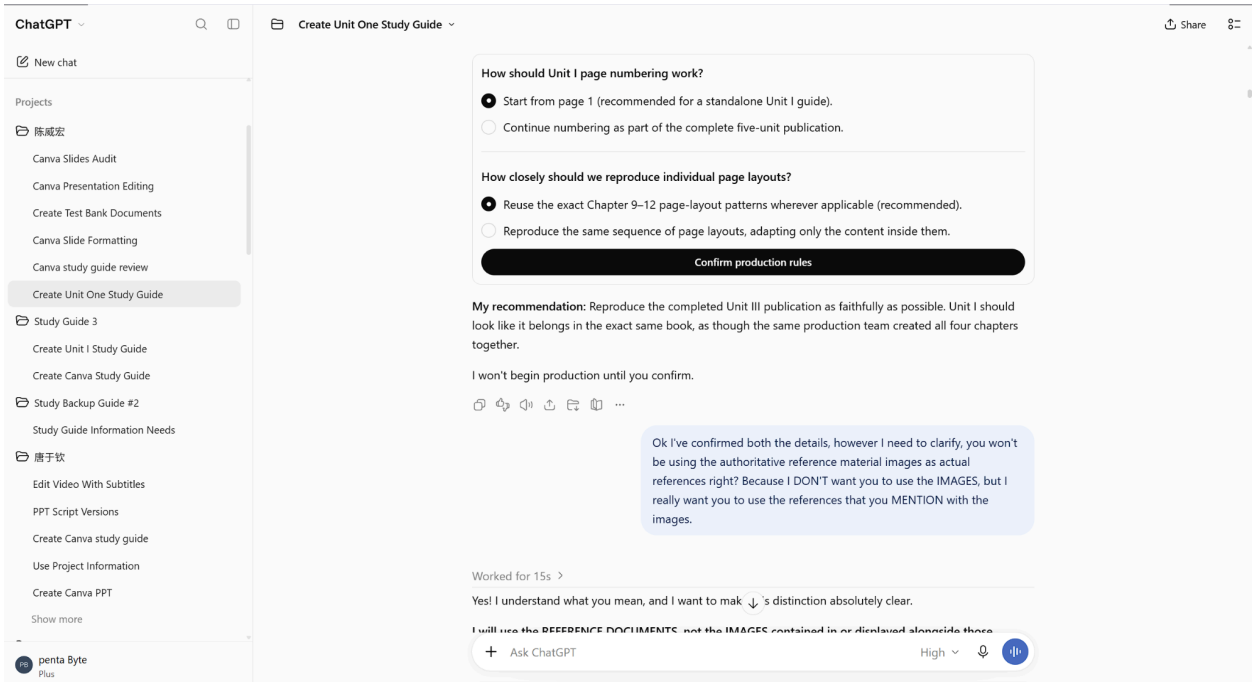


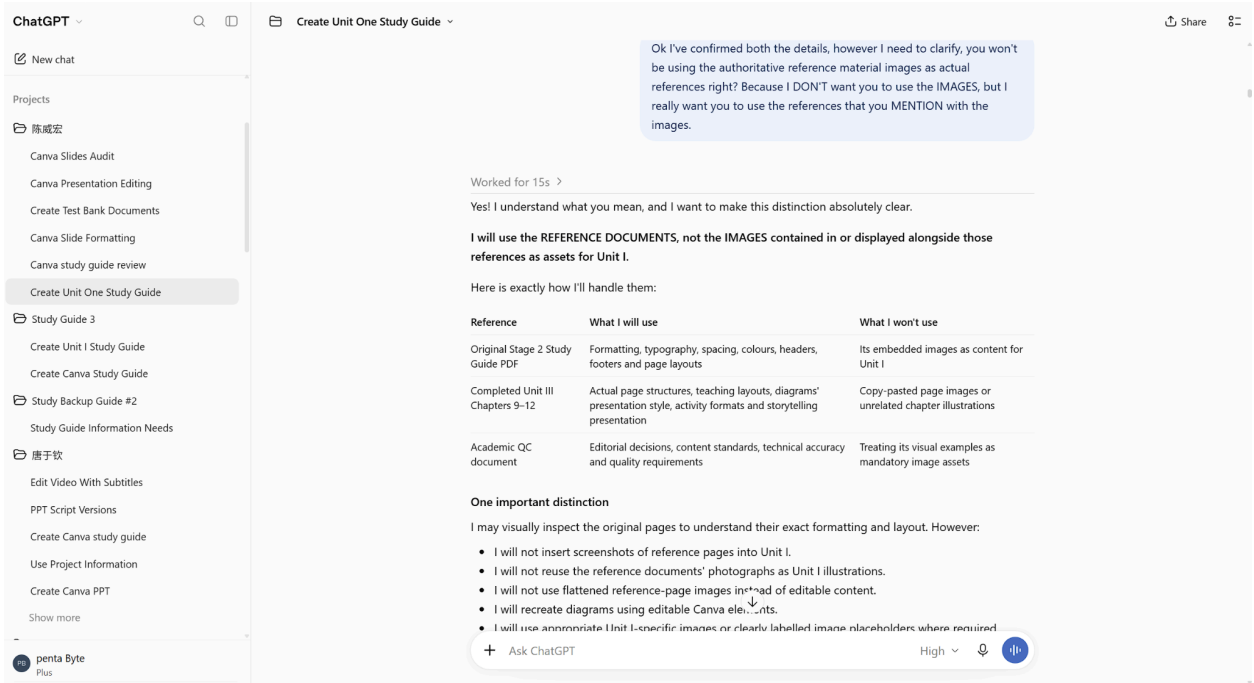
Reference A — Original Stage 2 Study Guide
`PentaByte_Stage2_StudyGuide(2).pdf`
Formatting authority
Pages 70–116, particularly Chapter 9. Establishes the original page dimensions, typography, margins, headers, footers, colours and layout system.

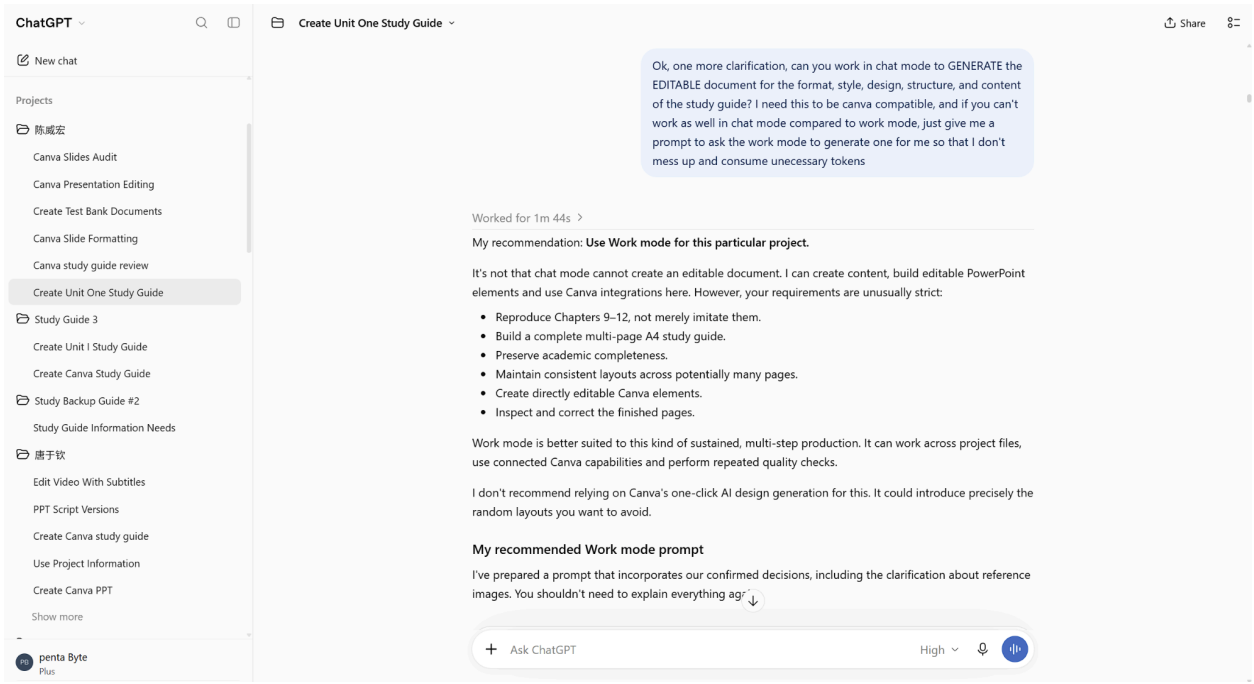


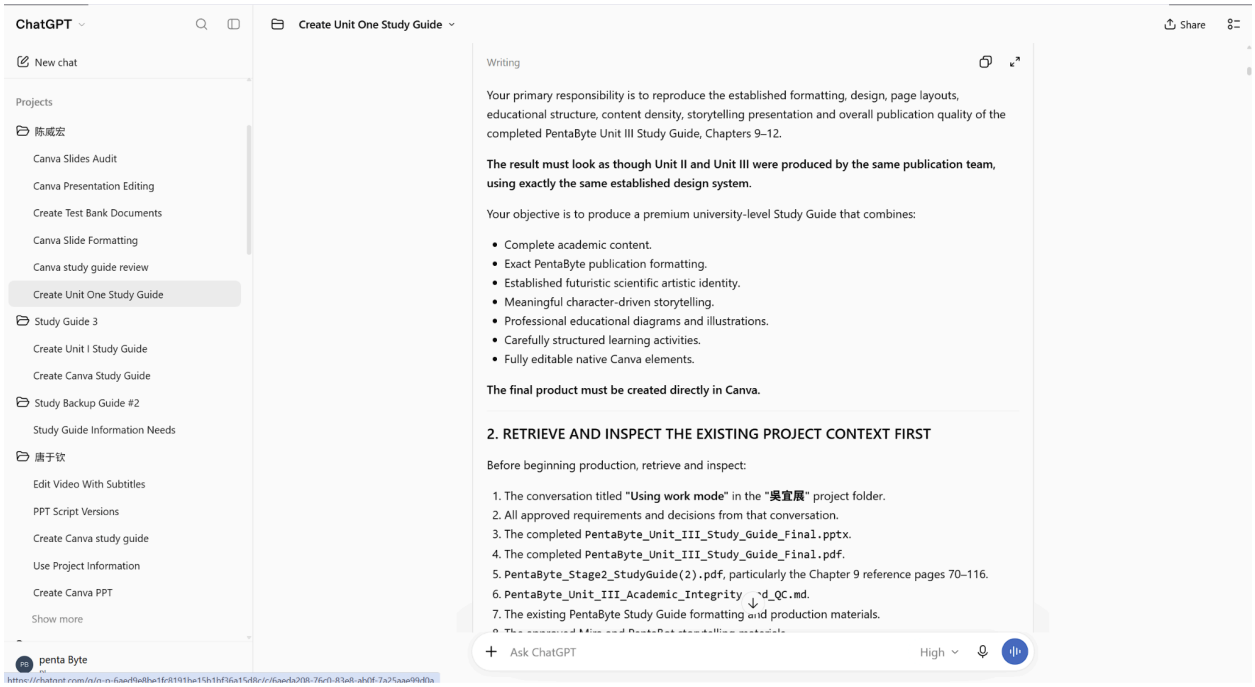
Reference B — Completed Unit III
`PentaByte_Unit_III_Study_Guide_Final.pptx` and PDF.
Complete production authority
All four chapters (9–12), including how the original design was extended across different types of explanations, diagrams, examples, activities and reviews.

Ask ChatGPT High


ChatGPT
New chat
Projects
陈威宏
Canva Slides Audit
Canva Presentation Editing
Create Test Bank Documents
Canva Slide Formatting
Canva study guide review
Create Unit One Study Guide
Study Guide 3
Create Unit I Study Guide
Create Canva Study Guide
Study Backup Guide #2
Study Guide Information Needs
唐于钦
Edit Video With Subtitles
PPT Script Versions
Create Canva study guide
Use Project Information
Create Canva PPT
Show more
penta Byte
Plus
Create Unit One Study Guide
Share
How should Unit I page numbering work?
Start from page 1 (recommended for a standalone Unit I guide).
Continue numbering as part of the complete five-unit publication.
How closely should we reproduce individual page layouts?
Reuse the exact Chapter 9–12 page-layout patterns wherever applicable (recommended).
Reproduce the same sequence of page layouts, adapting only the content inside them.
Confirm production rules
My recommendation: Reproduce the completed Unit III publication as faithfully as possible. Unit I should look like it belongs in the exact same book, as though the same production team created all four chapters together.
I won't begin production until you confirm.
Ok I've confirmed both the details, however I need to clarify, you won't be using the authoritative reference material images as actual references right? Because I DON'T want you to use the IMAGES, but I really want you to use the references that you MENTION with the images.
Worked for 15s
Yes! I understand what you mean, and I want to mak
s distinction absolutely clear.
Ask ChatGPT
High


ChatGPT
New chat
Projects
陈威宏
Canva Slides Audit
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Study Backup Guide #2
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唐于钦
Edit Video With Subtitles
PPT Script Versions
Create Canva study guide
Use Project Information
Create Canva PPT
Show more
penta Byte
Plus
Create Unit One Study Guide
Share
Ok I've confirmed both the details, however I need to clarify, you won't be using the authoritative reference material images as actual references right? Because I DON'T want you to use the IMAGES, but I really want you to use the references that you MENTION with the images.
Worked for 15s
Yes! I understand what you mean, and I want to make this distinction absolutely clear.
I will use the REFERENCE DOCUMENTS, not the IMAGES contained in or displayed alongside those references as assets for Unit I.
Here is exactly how I'll handle them:
Reference
What I will use
What I won't use
Original Stage 2 Study Guide PDF
Formatting, typography, spacing, colours, headers, footers and page layouts
Its embedded images as content for Unit I
Completed Unit III Chapters 9–12
Actual page structures, teaching layouts, diagrams' presentation style, activity formats and storytelling presentation
Copy-pasted page images or unrelated chapter illustrations
Academic QC document
Editorial decisions, content standards, technical accuracy and quality requirements
Treating its visual examples as mandatory image assets
One important distinction
I may visually inspect the original pages to understand their exact formatting and layout. However:
I will not insert screenshots of reference pages into Unit I.
I will not reuse the reference documents' photographs as Unit I illustrations.
I will not use flattened reference-page images instead of editable content.
I will recreate diagrams using editable Canva elements.
Ask ChatGPT
High





Created work prompt example:

MASTER WORK MODE PROMPT — PENTABYTE UNIT I STUDY GUIDE

1. YOUR ROLE

Act as the Lead Educational Content Designer, Academic Editor, Publication Designer and Canva Production Specialist for PentaByte.

Your task is to create the complete **Unit I Study Guide (Chapters 1–3)** as one professionally produced, fully editable, multi-page Canva design.

This is an existing publication project. You are NOT creating a new visual identity.

Your primary responsibility is to reproduce the established formatting, design, page layouts, educational structure, content density and presentation quality of the completed PentaByte Unit III Study Guide, Chapters 9–12.

The result must look as though Unit I and Unit III were produced by the same publication team, using the same established design system.

2. RETRIEVE THE EXISTING PROJECT CONTEXT FIRST

Before production, retrieve and examine:

1. The conversation titled **"Using work mode"** in the **"吳宜展"** project folder.
2. All relevant instructions and approved decisions from that conversation.
3. `PentaByte_Unit_III_Study_Guide_Final.pptx`.
4. `PentaByte_Unit_III_Study_Guide_Final.pdf`.
5. `PentaByte_Stage2_StudyGuide(2).pdf`, particularly printed pages 70–116.
6. `PentaByte_Unit_III_Academic_Integrity_and_QC.md`.
7. The latest approved Unit I Markdown files and content outlines.
8. Relevant Unit I notebooks, examples, assignment instructions and existing production materials.

Do not assume that the Unit III Markdown is the Unit I content source.

Identify the latest authoritative Unit I content and use it as the academic foundation.

If any source cannot be accessed, identify the exact missing file or information. Never pretend to have inspected unavailable material.

3. REFERENCE DOCUMENTS — IMPORTANT CLARIFICATION

I want you to use the REFERENCE DOCUMENTS, not their embedded images as assets.

The reference documents are authoritative for:

- Formatting.
- Page composition.
- Typography.
- Colour palette.
- Margins and spacing.
- Header and footer design.
- Chapter and subtopic layouts.
- Educational structure.
- Tables and diagrams' presentation style.
- Activity and checkpoint design.
- Storytelling presentation.
- Content density and explanation style.

You may inspect the original pages to understand these characteristics.

However:

- DO NOT copy and paste reference-page screenshots into Unit I.
- DO NOT reuse embedded photographs or illustrations from Chapters 9–12.
- DO NOT turn reference pages into flattened background images.
- DO NOT use the reference images as substitute content.
- Recreate diagrams using editable elements.
- Use appropriately labelled placeholders for required Unit I photographs, screenshots and other visuals.

The reference documents are authorities for the publication system, not image libraries.

4. NON-NEGOTIABLE DESIGN REQUIREMENT

Chapters 9–12 are the exact design and formatting benchmark.

I do not want a new design, a redesign, a new interpretation or a loosely inspired alternative.

Reproduce the established system:

- Same A4 portrait dimensions and proportions.
- Same page margins.
- Same typography and font hierarchy.
- Same chapter title and subtitle treatments.
- Same heading and subheading hierarchy.
- Same colours and colour combinations.
- Same header and footer bands.
- Same page-number positioning and styling.
- Same section dividers.
- Same chapter-opening layouts.
- Same explanation-page compositions.
- Same code-example presentation.
- Same diagram and table treatments.
- Same activity and learning-checkpoint layouts.
- Same takeaway and dialogue panels.
- Same references formatting.
- Same balance between text, visuals and whitespace.

Use the exact layout patterns from Chapters 9–12 wherever applicable.

When Unit I requires a particular type of page, find the closest equivalent in Chapters 9–12 and reproduce that layout.

Do not invent a new layout just because the academic topic is different.

Extract actual formatting specifications from the reference files rather than guessing their values.

5. UNIT I ACADEMIC CONTENT

Create the following three chapters:

CHAPTER 1 — From Problems to Algorithms

1.1 Identify inputs, expected output and smaller steps.

1.2 Write and trace a simple algorithm using pseudocode.

1.3 Work through the highest-score example and test unusual cases.

CHAPTER 2 — Essential Python for the Guide

2.1 Run a prepared notebook and understand variables, numbers, lists and output.

2.2 Read and modify conditions, loops and a simple function.

2.3 Complete a score-summary program and investigate how changing the pass mark affects results.

CHAPTER 3 — Measuring Algorithm Efficiency

1.1/3.1 Follow the approved Unit I numbering and content outline. Cover counting basic operations and understanding time complexity.

3.2 Recognize O(1), O(n) and O(n^2) visually, alongside any additional concepts specified in the approved content.

3.3 Compare time complexity and auxiliary space complexity.

IMPORTANT: The approved Unit I Markdown is the final authority for exact subtopic titles, explanations, examples and boundaries. Do not rewrite the approved scope.

Maintain exactly three numbered subtopics per chapter.

Preserve the academic content, including:

- Complete explanations.
- Definitions.
- Worked examples.
- Pseudocode.
- Python code.
- Trace tables.
- Expected outputs.
- Edge cases.
- Activities.
- Prediction questions.
- Answers and feedback.
- Chapter takeaways.
- Learning checkpoints.
- References.

Do not oversimplify, remove explanations or condense material simply to reduce the page count.

6. STORYTELLING — MIRA AND PENTABOT

Maintain the educational storyline established for PentaByte.

Mira:

- Curious beginner.
- Asks meaningful questions.
- Makes observations and mistakes.
- Develops understanding progressively.
- Makes decisions and applies what she learns.

PentaBot:

- Intelligent, friendly and occasionally humorous.
- Challenges assumptions.
- Gives hints rather than immediately revealing answers.
- Corrects misconceptions.
- Encourages independent reasoning.

The story must develop continuously across Chapters 1–3.

Use the same dialogue-panel design and presentation approach established in Chapters 9–12.

IMPORTANT: Do not include pictures of PentaBot. Textual references and dialogue are permitted.

7. EDITABILITY AND CANVA COMPATIBILITY

The final product must be a genuine, editable Canva design.

Use native editable elements wherever possible:

- Text.
- Shapes.
- Lines.
- Tables.
- Flowcharts.
- Diagrams.
- Code panels.
- Learning panels.
- Labels.

- Headers and footers.
- Page numbers.

Do not flatten complete pages into images.

Do not deliver a PDF or PowerPoint as a substitute for the requested Canva project.

If an intermediate PowerPoint is necessary, it must contain native editable elements and must ultimately be imported into Canva and checked for editability.

Use the actual Canva integration or supported browser workflow. Do not claim to have created a Canva project without a working editing link.

8. PAGE COUNT AND PAGINATION

- A4 portrait.
- Start Unit I pagination from page 1.
- No predetermined page count.
- Let academic content and teaching difficulty determine the number of pages.
- Follow the content density and educational pacing of Chapters 9–12.
- Do not artificially lengthen simple topics.
- Do not compress difficult topics.
- Give every page a clear educational purpose.

9. PRODUCTION WORKFLOW

Follow these stages:

Phase 1 — Reference audit

Inspect Chapters 9–12 and extract the existing publication system.

Phase 2 — Academic audit

Cross-reference the complete Unit I content against the latest approved Markdown, assignment requirements and supplied notebooks.

Phase 3 — Page planning

Map every approved content section to an appropriate existing Chapter 9–12 layout.

Do not create new design patterns unnecessarily.

Phase 4 — Canva construction

Build the complete Unit I guide using editable Canva elements.

Preserve the original Unit III files and designs.

Phase 5 — Academic verification

Check:

- Every definition.
- Every explanation.
- Every Python code example.
- Every expected output.
- Every calculation.
- Every activity and answer.
- Every technical statement.
- Every reference.

Execute Python examples where appropriate to verify their outputs.

Phase 6 — Visual quality control

Review every page for:

- Consistency with Chapters 9–12.
- Text clipping.
- Overlapping elements.
- Alignment.
- Font consistency.
- Spacing.
- Page numbering.
- Diagram accuracy.
- Readability.
- Editability.

Correct identified problems before delivery.

10. FINAL DELIVERABLE

Produce ONE complete project:

PentaByte — Unit I Study Guide

It must contain all three chapters and the necessary opening, review, glossary and reference sections.

Final delivery requirements:

1. One working Canva editing link.
2. A complete, professionally formatted and editable Unit I Study Guide.
3. Confirmation that the academic and visual quality checks were performed.

Do not deliver an incomplete draft and describe it as final.

Do not create unrelated reports, alternative designs or unnecessary files.

11. FINAL INSTRUCTION

The most important instruction throughout this project:

REPRODUCE THE EXISTING PENTABYTE UNIT III PUBLICATION SYSTEM FOR UNIT I.

I do not want your own interpretation of what a professional study guide should look like.

I want the same formatting, same design language, same educational structure, same page-layout patterns, same content density and same presentation quality already achieved in Chapters 9–12.

Only the academic subject matter changes.

Do not ask me to repeat instructions already established in the project. Use your judgment for minor production decisions, but never override approved design or academic requirements.

Begin by auditing the source materials, then proceed with production.

Next Steps

Finalize every deliverable